

HTTP :30080 (NodePort)

KUBERNETES CLUSTER · namespace: medimatrix

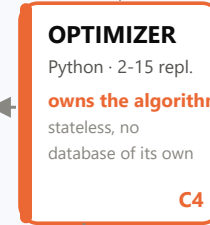
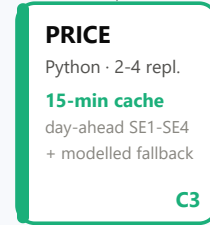
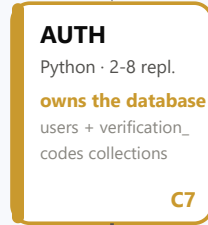
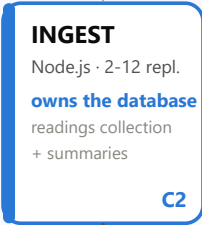
16 NetworkPolicy objects: default-deny-all plus 15 explicit allows (DNS + one pair per legitimate call). ClusterIP everywhere except the gateway.

API GATEWAY

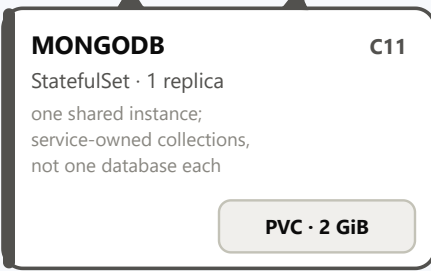
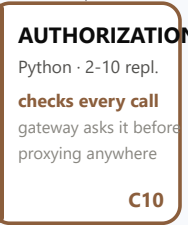
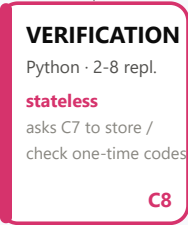
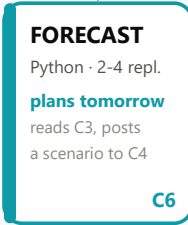
C1

Node.js / Express · 2-10 replicas
only externally reachable service · routing, rate limit, security headers

TIER 1 · owns something



TIER 2 · reads only, builds on Tier 1



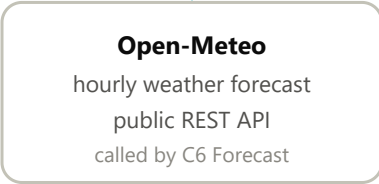
mongodb :27017 — only pods labelled
app:ingest or app:auth may connect
(enforced by NetworkPolicy, not just
convention)

optimizer → price, ingest (internal REST, over Kubernetes DNS)

HTTPS

HTTPS

EXTERNAL · outside the cluster, HTTPS, no API key



Solid = request path · Dashed = internal service-to-service REST calls · Only C1 is reachable from outside the cluster · Only C3 and C6 may reach the public internet · Only C2 and C7 may reach MongoDB
Roles are self-declared at registration: "staff" (read + write) or "viewer" (read-only), enforced by C10 · Docker Desktop's default CNI does not enforce the NetworkPolicy objects shown above
Not drawn for legibility: assistant (C5) also reads C3 and C4; forecast (C6) also reads C4; validation (C9) also reads C2; verification (C8) and auth (C7) call each other. Full call graph: README.md / docs/01-REPORT.md.